

ARTICLE NAVIGATION

NEW STRATEGIES FOR EARLY PHASE CLINICAL TRIALS | SEPTEMBER 05 2024

Abstract B071: Automated extraction and provision of electronic health record data from children with cancer to National Childhood Cancer Center (NCCR) cancer registries

FREE

Tamara P. Miller; Johanna Goderre Jones; Elizabeth Hsu; Edward Krause; Yun Gun Jo; Judy Lee; Kevin Ward; Lynne Penberthy; Richard Aplenc



+ [Author & Article Information](#)

Cancer Res (2024) 84 (17_Supplement): B071.

<https://doi.org/10.1158/1538-7445.PEDIATRIC24-B071>

 Split-Screen

 Share ▾

 Tools ▾

 Versions ▾

Abstract

Introduction: The National Childhood Cancer Registry (NCCR) is a central infrastructure that integrates childhood cancer data from central cancer registries and other sources to enhance access to and utilization of pediatric cancer data. The NCCR currently represents >70% of the United States pediatric population. Registries collect and publish cancer incidence and survival data for public health surveillance. Electronic health records (EHRs) are one data source. EHR data are currently obtained by manual collection at individual hospitals that then transfer data to population-based cancer registries. This study aimed to use automated data collection from hospital EHRs to broaden and standardize data reported to registries. **Methods:** ExtractEHR is an R based software tool that extracts and formats data derived from EHRs from multiple institutions and vendors for research initiatives. ExtractEHR was installed by the hospital technical team to extract the following EHR components: addresses; demographics; inpatient and clinic visits; laboratory, microbiology, pathology, genomic, and radiology test results; medications orders and administrations; procedures; height and weight; and oncology clinician notes. Patients were identified by the central cancer registry and case lists provided to the hospital team. ExtractEHR extracted data for all patients from time of diagnosis through most recent encounter in the EHR. Data elements were extracted as comma-separated values (csv) files. The hospital clinical lead checked data for completeness and worked with the hospital technical and ExtractEHR teams to iteratively adjust ExtractEHR mapping to ensure complete data extraction. Once considered complete, csv files were transferred to the registry team using a secure file transfer system. The initial pilot was completed for patients with 5

diagnosis types over a 2-year period at Children's Healthcare of Atlanta (CHOA) and transferred to the Georgia Cancer Registry. Results: Data from 306 patients with astrocytoma, Hodgkin Lymphoma, neuroblastoma, Ewing or osteosarcoma, or leukemia diagnosed at CHOA in 2019 or 2020 were extracted using ExtractEHR. There were 3,015,822 data elements extracted across all EHR components, including 832,006 laboratory test results, 574,462 medication administrations, and 34,756 oncology notes. Both structured data (e.g. laboratory results) and unstructured free text data (e.g. notes) were successfully extracted. Data stored in PDFs in the EHR were not extracted due to storage parameters in the EHR data warehouse. All extracted data were successfully transferred to the registry. Conclusions: ExtractEHR can extract and transfer a broad range of clinical data on children treated for cancer to cancer registries. The data granularity obtained using ExtractEHR is impossible to obtain manually. Work is ongoing to expand to all cancers and implement ExtractEHR at 4 additional hospitals. These data will be deidentified by the registry teams and submitted to the NCCR to expand the comprehensiveness of clinical data included in the NCCR.

Citation Format: Tamara P. Miller, Johanna Goderre Jones, Elizabeth Hsu, Edward Krause, Yun Gun Jo, Judy Lee, Kevin Ward, Lynne Penberthy, Richard Aplenc. Automated extraction and provision of electronic health record data from children with cancer to National Childhood Cancer Center (NCCR) cancer registries [abstract]. In: Proceedings of the AACR Special Conference in Cancer Research: Advances in Pediatric Cancer Research; 2024 Sep 5-8; Toronto, Ontario, Canada. Philadelphia (PA): AACR; Cancer Res 2024;84(17 Suppl):Abstract nr B071.

©2024 American Association for Cancer Research

Advertisement

[Skip to Main Content](#)

[View Metrics](#)

Citing Articles Via

[Google Scholar](#)

Email Alerts

[Article Activity Alert](#)

[Online First Alert](#)

[eTOC Alert](#)

[Editors' Picks Alert](#)

Online ISSN 1538-7445 **Print ISSN** 0008-5472

AACR Journals

[Skip to Main Content](#)



- Blood Cancer
Discovery

Cancer Research
Communications
- Cancer
Epidemiology,
Biomarkers &
Prevention

Clinical Cancer
Research
Molecular Cancer
Research
- Cancer Immunology
Research

Molecular Cancer
Therapeutics
- Cancer Prevention
Research

Resources

- Information on Advertising & Reprints

Information for Institutions/Librarians

RSS Feeds

Privacy Policy

AACR Journals on Social Media



Copyright © 2026 by the American Association for Cancer Research.